

ECMT676 Econometrics II
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Tentative Solutions to Problem 3

1. (7.2)

(a) Since $\sqrt{N}(\hat{\beta}_{OLS} - \beta) = (N^{-1} \sum X_i' X_i)^{-1} (N^{-1/2} \sum X_i' u_i)$, then

$$AVar(\hat{\beta}_{OLS}) = [E(X_i' X_i)]^{-1} [E(X_i' \Omega X_i)] [E(X_i' X_i)]^{-1} / N.$$

(b) We can estimate $E(X_i' X_i)$ by $N^{-1} \sum X_i' X_i$ and $E(X_i' \Omega X_i)$ by $N^{-1} \sum X_i' \hat{\Omega} X_i$, where $\hat{\Omega} = N^{-1} \sum \hat{u}_i \hat{u}_i'$ and where $\hat{u}_i = Y_i - X_i \hat{\beta}_{OLS}$.

(c) Note that $AVar(\hat{\beta}_{FGLS}) = N^{-1} E(X_i' \Omega^{-1} X_i)^{-1}$. Then

$$\begin{aligned} AVar(\hat{\beta}_{FGLS})^{-1} - AVar(\hat{\beta}_{OLS})^{-1} &= N \left[E(X_i' \Omega^{-1} X_i) - E(X_i' X_i) [E(X_i' \Omega X_i)]^{-1} E(X_i' X_i) \right] \\ &= N \left[E(A_i' A_i) - E(A_i' B_i) [E(B_i' B_i)]^{-1} E(B_i' A_i) \right] \\ &= N E(C_i' C_i) \end{aligned}$$

where $A_i = \Omega^{-1/2} X_i$, $B_i = \Omega^{1/2} X_i$, and $C_i = A_i - B_i [E(B_i' B_i)]^{-1} E(B_i' A_i)$.

(d) If $\Omega = \sigma^2 \mathbf{I}_G$, then $AVar(\hat{\beta}_{OLS}) = N^{-1} \sigma^2 E(X_i' X_i)^{-1}$.

(e) Yes, the statement is correct.

2. (7.14) First, we can derive the asymptotic variance of $\hat{\beta}_{FGLS}$ which is

$$AVar(\hat{\beta}_{FGLS}) = [E(X_i' \Lambda^{-1} X_i)]^{-1} E(X_i' \Lambda^{-1} \Omega \Lambda^{-1} X_i) [E(X_i' \Lambda^{-1} X_i)]^{-1}.$$

We want to prove that $AVar(\hat{\beta}_{FGLS}^{\hat{\Lambda}}) - AVar(\hat{\beta}_{FGLS}^{\hat{\Omega}})$ is positive semi-definite. We can rewrite that

$$\begin{aligned} AVar(\hat{\beta}_{FGLS}^{\hat{\Omega}})^{-1} - AVar(\hat{\beta}_{FGLS}^{\hat{\Lambda}})^{-1} &= N \left[E(X_i' \Omega X_i) - E(X_i' \Lambda^{-1} X_i) [E(X_i' \Lambda^{-1} \Omega \Lambda^{-1} X_i)]^{-1} E(X_i' \Lambda^{-1} X_i) \right] \\ &= N \left[E(\tilde{X}_i' \tilde{\Omega}^{-1} \tilde{X}_i) - E(\tilde{X}_i' \tilde{X}_i) [E(\tilde{X}_i' \tilde{\Omega} \tilde{X}_i)]^{-1} E(\tilde{X}_i' \tilde{X}_i) \right] \end{aligned}$$

where $\tilde{X}_i = \Lambda^{-1/2} X_i$ and $\tilde{\Omega} = \Lambda^{1/2} \Omega \Lambda^{1/2}$. Following the similar trick, from 7.2 (c), we can have the desired result.

3. (8.8)

(a) First we can write the panel data model as $Y_i = X_i' \beta + u_i$ where $Y_i = (y_{i1}, \dots, y_{iT})'$ and $X_i =$

$(x_{i1}, \dots, x_{iT})$, and then¹

$$\begin{aligned}
\hat{\beta}_{\text{S2SLS}} &= \left(\sum_i \hat{X}_i \hat{X}_i' \right)^{-1} \sum_i \hat{X}_i Y_i \\
&= \left(\sum_i X_i' Z_i \left(\sum_i Z_i' Z_i \right)^{-1} \sum_i Z_i' X_i \right)^{-1} \left(\sum_i X_i Z_i' \left(\sum_i Z_i' Z_i \right)^{-1} \sum_i Z_i Y_i \right) \\
&= \left(\sum_i \sum_t x_{it} z_{it}' \left(\sum_i \sum_t z_{it} z_{it}' \right)^{-1} \sum_i \sum_t z_{it} x_{it}' \right)^{-1} \\
&\quad \left(\sum_i \sum_t x_{it} z_{it}' \left(\sum_i \sum_t z_{it} z_{it}' \right)^{-1} \sum_i \sum_t z_{it} y_{it} \right) \\
&= \hat{\beta}_{\text{P2SLS}}
\end{aligned}$$

- (b) $E(\sum_t z_{it} x_{it}') = K$ and $E(\sum_t z_{it} z_{it}') = L$.
(c) First, from (b), we can prove that the asymptotic variance of $\hat{\beta}_{\text{P2SLS}}$ is $N^{-1} A^{-1} B A^{-1}$ where

$$\begin{aligned}
A &= E(X_i Z_i') (E(Z_i Z_i'))^{-1} E(Z_i X_i') \\
B &= E(X_i Z_i') (E(Z_i Z_i'))^{-1} E(Z_i u_i u_i' Z_i') (E(Z_i Z_i'))^{-1} E(Z_i X_i').
\end{aligned}$$

Then, estimate $E(X_i Z_i')$ by $N^{-1} \sum_i X_i Z_i'$, $E(Z_i Z_i')$ by $N^{-1} \sum_i Z_i Z_i'$, and $E(Z_i u_i u_i' Z_i')$ by $N^{-1} \sum_i Z_i \hat{u}_i \hat{u}_i' Z_i'$, where $\hat{u}_i = Y_i - X_i' \hat{\beta}_{\text{2SLS}}$.

- (d) Since $E(Z_i u_i) = 0$, then the optimal weighting matrix is $\hat{S}^{-1}$ where $\hat{S} = N^{-1} \sum_i Z_i \hat{u}_i \hat{u}_i' Z_i'$ and $\hat{u}_i = Y_i - X_i' \hat{\beta}_{\text{2SLS}}$.
(e) Note that $E(Z_i u_i u_i' Z_i') = E\left(\sum_{s=1}^T \sum_{t=1}^T z_{is} u_{is} u_{it} z_{it}'\right) = \sum_{s=1}^T \sum_{t=1}^T E(z_{is} u_{is} u_{it} z_{it}')$. Then, with equation (8.65) and (8.66), we can prove that $E(z_{it} u_{it}^2 z_{it}') = E(z_{it} E[u_{it}^2 | z_{it}] z_{it}') = \sigma^2 E(z_{it} z_{it}')$ and $E(z_{is} u_{is} u_{it} z_{it}') = E[z_{is} u_{is} E(u_{it} | u_{ik}, z_{ik}, k \leq t) z_{it}'] = 0$ provided $s < t$.² Thus, $E(Z_i u_i u_i' Z_i') = \sigma^2 \sum_t E(z_{it} z_{it}') = \sigma^2 E(Z_i Z_i')$. Then, $\text{AVar}(\hat{\beta}_{\text{P2SLS}}) = N^{-1} \sigma^2 E(X_i Z_i') (E(Z_i Z_i'))^{-1} E(Z_i X_i')$.
(f) If $E(u_{it}^2 | z_{it}) = E(u_{it}^2) \equiv \sigma_t^2$, $t = 1, \dots, T$. Then, $E(Z_i u_i u_i' Z_i') = E(Z_i \Omega Z_i')$ where Ω is a diagonal matrix whose diagonal is $\sigma_1^2, \dots, \sigma_T^2$. Now, we can use 3SLS.
(g) We denote that $Z_i = \text{diag}(z_{i1}, \dots, z_{iT})$ which is a $\sum L \times T$ matrix. Then, $\hat{X}_i' = Z_i' (\sum_i Z_i Z_i')^{-1} \sum_i Z_i X_i'$. Thus,

$$\begin{aligned}
\hat{\beta}_{\text{IV}} &= \left(\sum_i \hat{X}_i \hat{X}_i' \right)^{-1} \sum_i \hat{X}_i Y_i \\
&= \left(\sum_i X_i Z_i' \left(\sum_i Z_i Z_i' \right)^{-1} \sum_i Z_i X_i' \right)^{-1} \sum_i X_i Z_i' \left(\sum_i Z_i Z_i' \right)^{-1} \sum_i Z_i Y_i \\
&= \hat{\beta}_{\text{S2SLS}}
\end{aligned}$$

- (h) Generalized Moment Method.

4. **(10.10)**
5. **(11.16)**

¹Here, we denote $Z_i = (z_{i1}, \dots, z_{iT})$ a $L \times T$ matrix which is different Wooldridge's notation.

²Following the similar argument, we can prove $E(z_{is} u_{is} u_{it} z_{it}') = 0$ provided $t < s$.

- (a) We can estimate β by FE approach.
(b)

$$\begin{aligned}
E(y_{it}|w_i, x_i, r_i) &= E(\gamma w_i + x_{it}\beta + c_i + u_{it}|w_i, x_i, r_i) \\
&= \gamma w_i + x_{it}\beta + E(c_i|w_i, x_i, r_i) + E(u_{it}|w_i, x_i, r_i) \\
&= \gamma w_i + x_{it}\beta + \delta_0 + \delta_1 r_i + \bar{x}_i \delta_2 + E(E(u_{it}|w_i, x_i, r_i, c_i)|w_i, x_i, r_i) \\
&= \gamma w_i + x_{it}\beta + \delta_0 + \delta_1 r_i + \bar{x}_i \delta_2
\end{aligned}$$

- (c) Provided a standard rank condition holds for the explanatory variables, γ is identified because it appears in a conditional expectation containing observable variables: $E(y_{it}|w_i, x_i, r_i)$.
(d) $y_{it} = \gamma w_i + x_{it}\beta + c_i + u_{it} = \gamma w_i + x_{it}\beta + \delta_0 + \delta_1 r_i + \bar{x}_i \delta_2 + a_i + u_{it}$. Denote $\varepsilon_{it} = a_i + u_{it}$. Then, we can prove that $\text{Var}(\varepsilon_{it}|w_i, x_i, r_i) = \text{Var}(a_i + u_{it}|w_i, x_i, r_i) = \text{Var}(a_i|w_i, x_i, r_i) + \text{Var}(u_{it}|w_i, x_i, r_i) = \sigma_a^2 + \sigma_u^2$ and $\text{Cov}(\varepsilon_{it}, \varepsilon_{is}|w_i, x_i, r_i) = \sigma_a^2$.³ Thus, we conclude that $\text{Var}(\varepsilon_i)$ has the random effects form, and so we should just apply the usual REGLS estimator. This is asymptotically more efficient than the pooled OLS estimator.

³ $E(a_i u_{is}|w_i, x_i, r_i) = E\{[c_i - E(c_i|w_i, x_i, r_i)]u_{is}|w_i, x_i, r_i\} = E(c_i u_{is}|w_i, x_i, r_i) - E(c_i|w_i, x_i, r_i)E(u_{it}|w_i, x_i, r_i) = E(c_i E(u_{is}|w_i, x_i, r_i, c_i)|w_i, x_i, r_i) - E(c_i|w_i, x_i, r_i)E(E(u_{it}|w_i, x_i, r_i, c_i)|w_i, x_i, r_i) = 0$